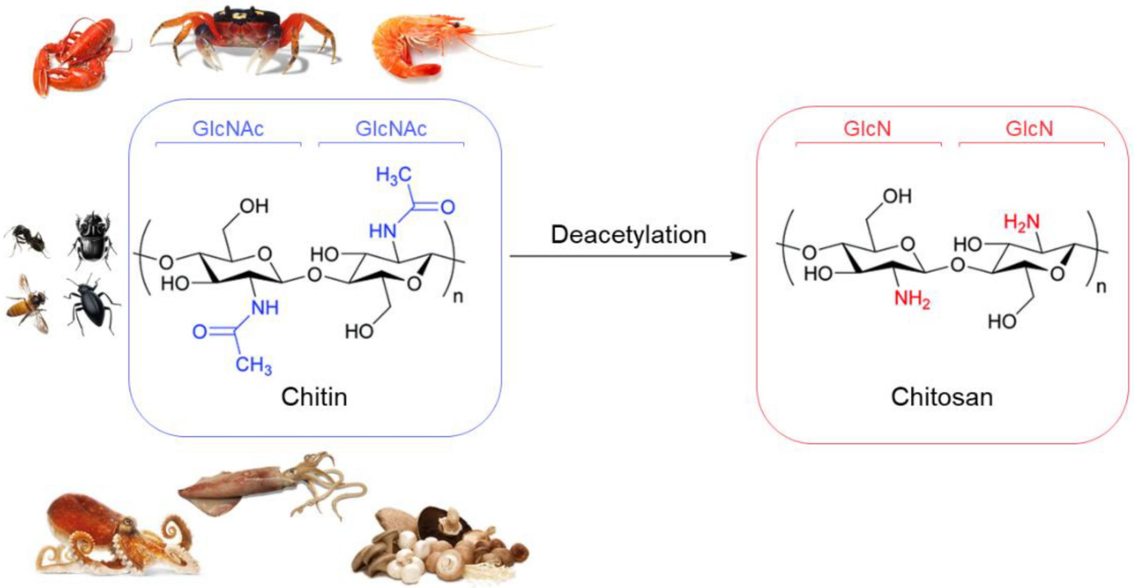
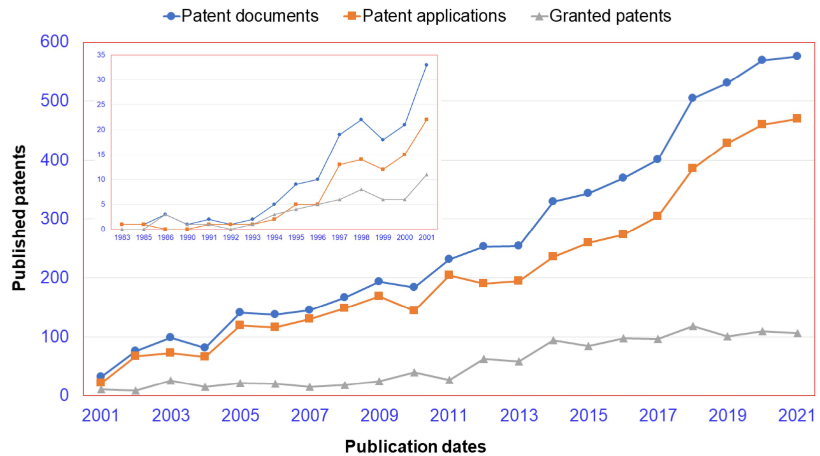
Hydrogels

Natural Hydrogels

Chitosan: Cationic Natural Hydrogel Former (I)

It is a (partially) deacetylated derivative of chitin, the second most abundant polymer on Earth. Chitin is insoluble!





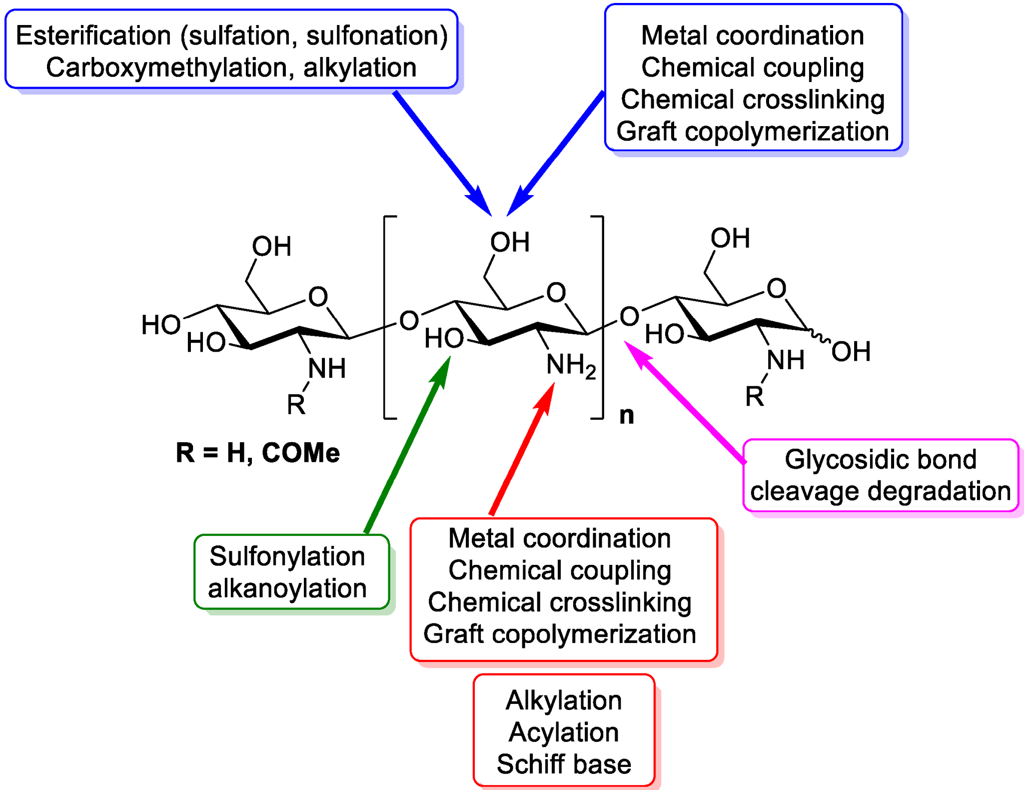
Chitin and Chitosan-Based Hydrogel and Their Use in the Agriculture
In: Bachheti, R.K., Bachheti, A., Husen, A. (eds) Chitin-Based Nanoparticles for the Agriculture Sectors. Smart Nanomaterials Technology. Springer, Singapore.
https://doi.org/10.1007/978-981-96-0920-8_12

Mater. Proc. **2022**, *9*, 1

Hydrogels

Natural Hydrogels

Chitosan: Cationic Natural Hydrogel Former (II)



Polymers **2021**, *13*, 3256

Int. J. Biol. Macromol. **2024**, *276*, 133823

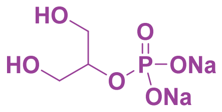
Synthesis method	Type	Process	Properties
Physical cross-linking	Ionic Gelation	Mixing chitosan with polyanionic compounds like TPP (tripolyphosphate)	Biocompatible, non-toxic, easy to form, limited mechanical strength
	Freeze-Thaw Cycles	Repeated cycles of freezing and thawing the chitosan solution	Improved mechanical strength, porous structure, better water absorption
	pH Induced Gelation	Adjusting the pH of chitosan solution to form a gel	Simple process, pH-sensitive, useful for drug delivery
Chemical cross-linking	Covalent Bonding	Using cross-linkers like glutaraldehyde, genipin, or EDC (1-Ethyl-3-(3-dimethylaminopropyl) carbodiimide)	Stronger mechanical properties, tunable degradation rates, potential cytotoxicity
	Enzymatic Cross-linking	Using enzymes like transglutaminase to cross-link chitosan	Mild reaction conditions, biocompatible, slower reaction rate
Radiation cross-linking	Gamma Irradiation	Exposing chitosan to gamma rays to induce cross-linking	Sterile, no need for chemical cross-linkers, potential for structural damage
	UV Irradiation	Using UV light with a photo-initiator to cross-link chitosan	Controlled reaction, suitable for surface modification, limited penetration depth
Thermal cross-linking	Heat Treatment	Heating chitosan in the presence of cross-linkers or by itself	Enhanced mechanical strength, thermal stability, risk of degradation

Chemical crosslinking

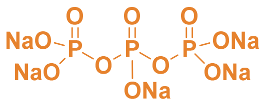
Natural Hydrogels

Chitosan: Cationic Natural Hydrogel Former (III)

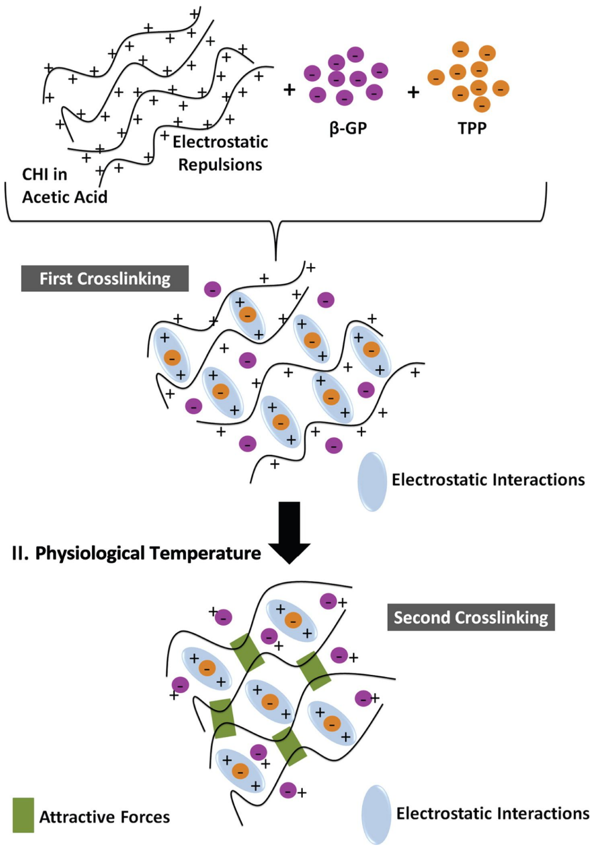
Ionic crosslinking



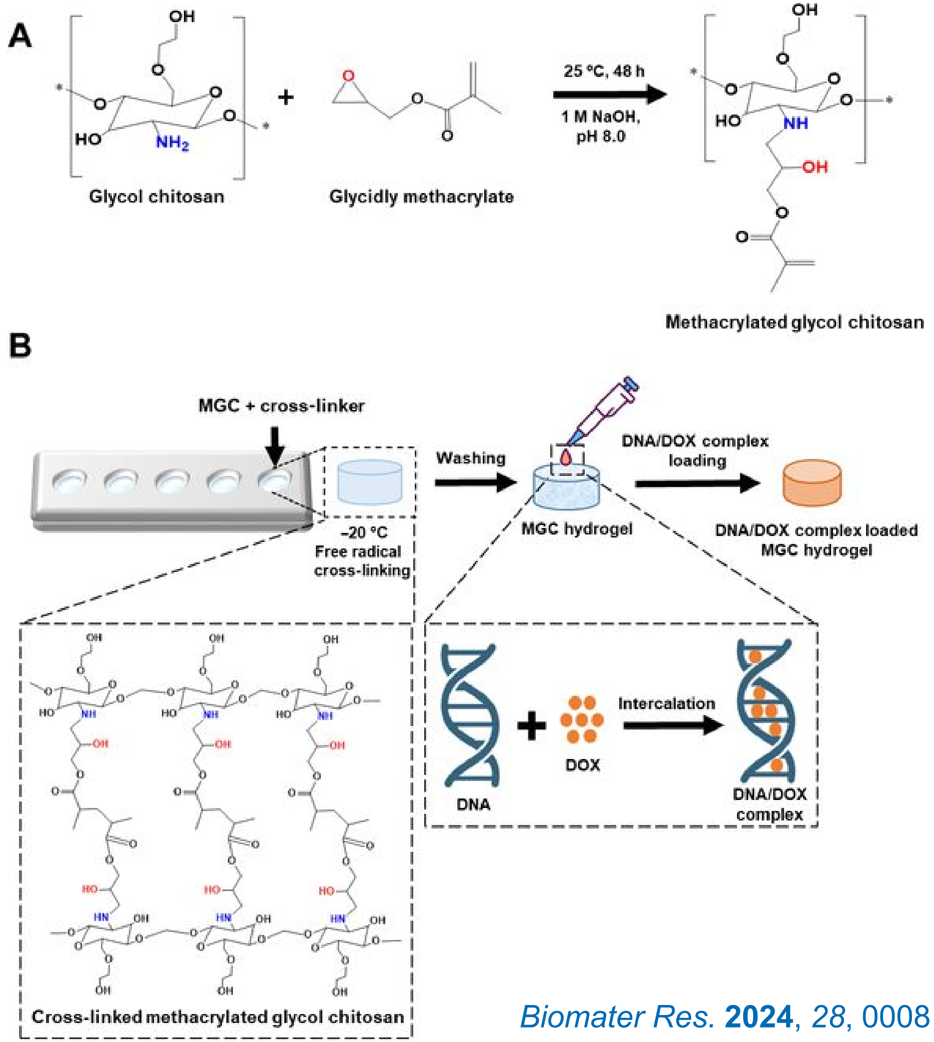
β-glycerophosphate



tripolyphosphate



J. Bioact. Compat. Polym. **2014**, *29*, 50



Biomater Res. **2024**, *28*, 0008

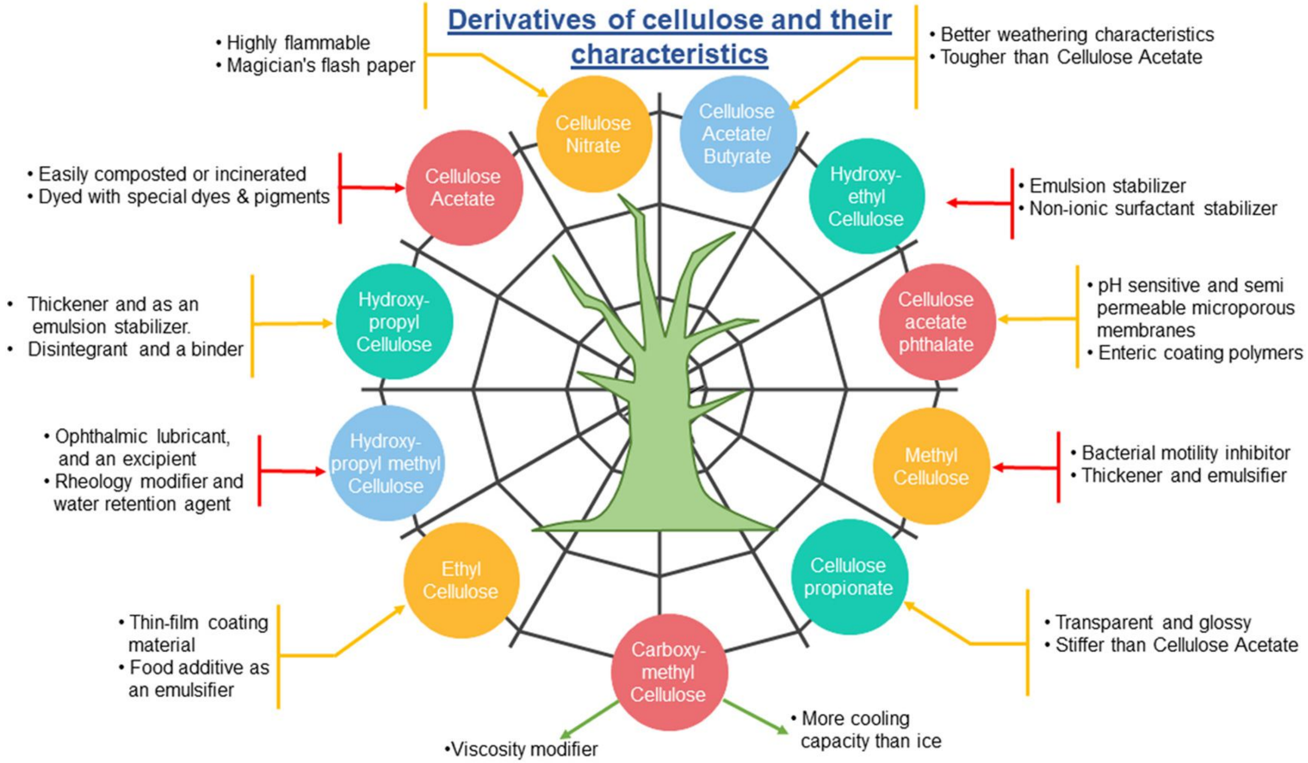
Int. J. Biol. Macromol. **2019**, *129*, 750

Natural Hydrogels

Cellulose-Based Hydrogels (I)

Most abundant polymer on Earth but insoluble in all common solvents*!

To make hydrogels, cellulose needs to solubilized and therefore be modified (see viscose process):



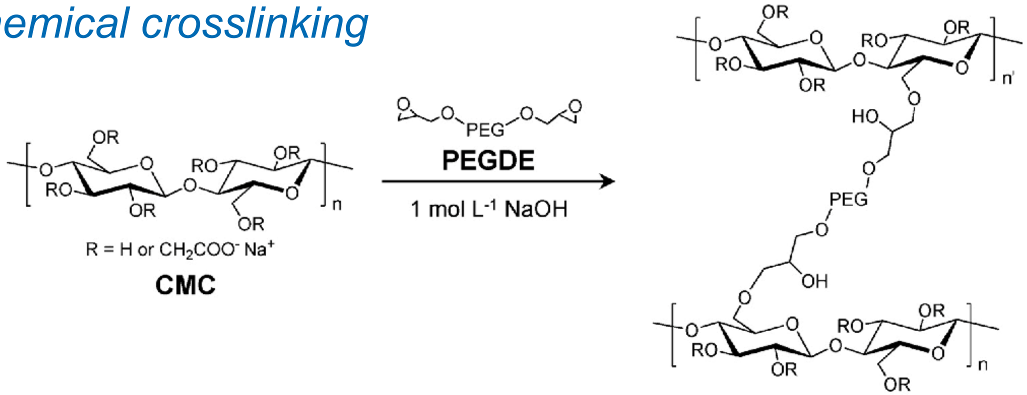
*examples of solvent systems : *N*-methylmorpholine oxide, LiCl in DMAc or NMP, paraformaldehyde/DMSO, tetralkylammonium salt in DMSO

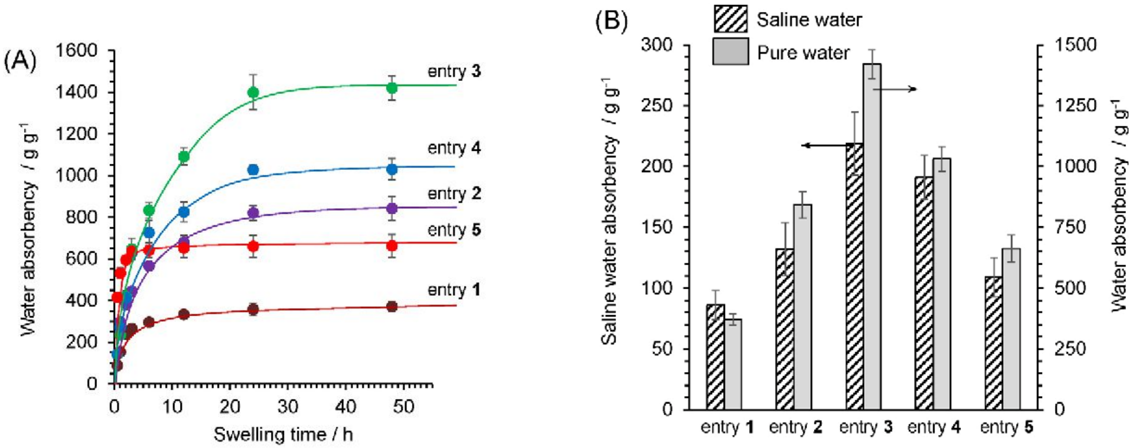
Hydrogels

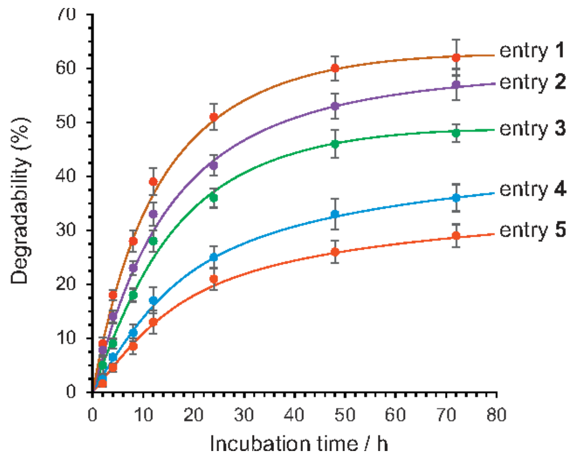
Natural Hydrogels

Cellulose-Based Hydrogels (II)

Chemical crosslinking







Carbohydr. Polym. **2014**, *106*, 84

Hydrogels

Synthetic Hydrogels

General Aspects

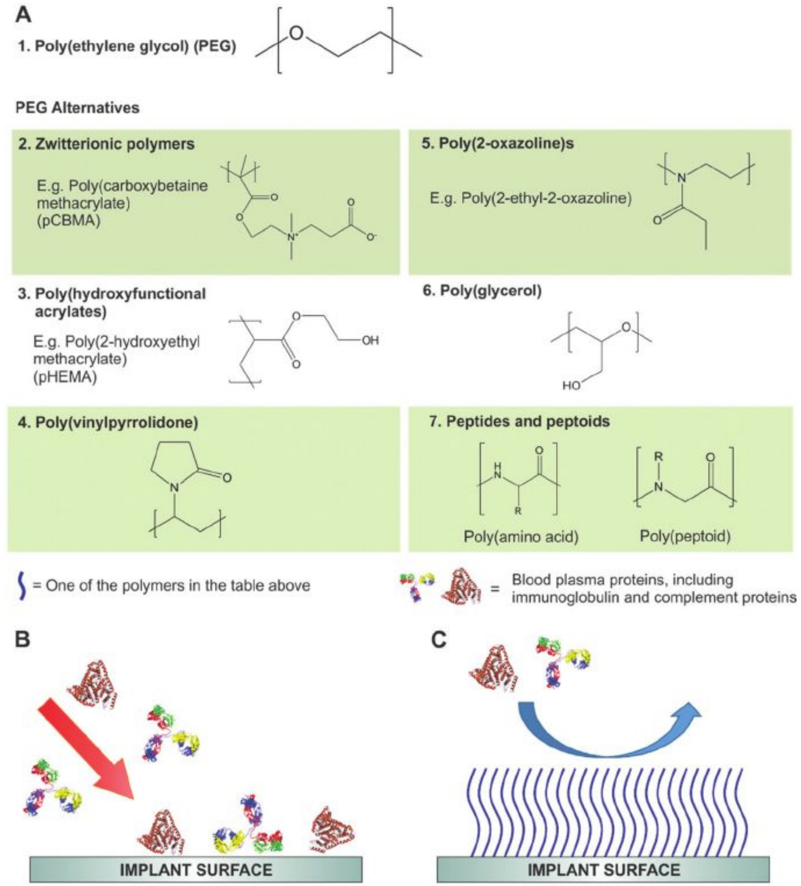
Unlike natural hydrogels, synthetic systems can be designed with ***more***:

- controlled crosslink density
- defined monomer composition
- reproducible network architecture
- tunable functionality

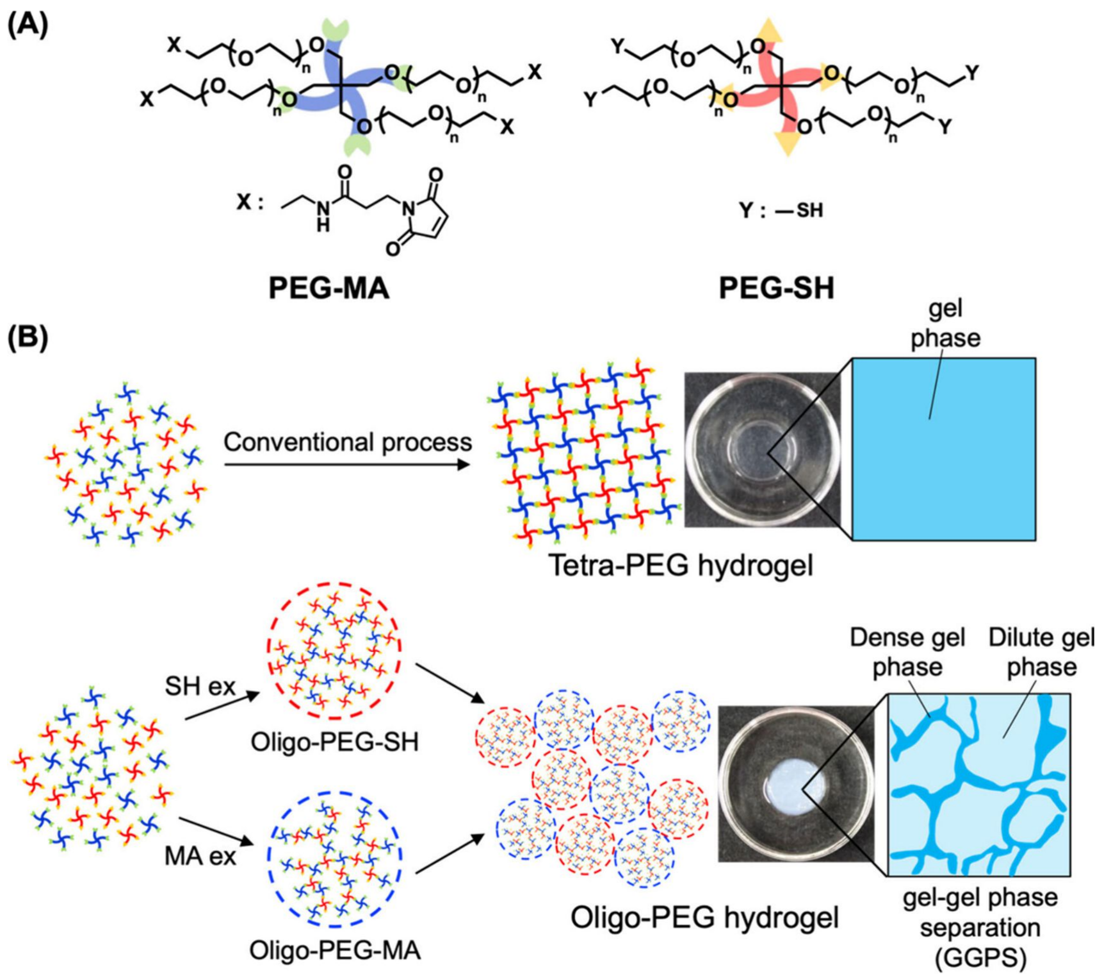
Hydrogels

Synthetic Hydrogels

PEG Hydrogels: The Bioinert Standard



Polym. Chem. **2015**, *6*, 198

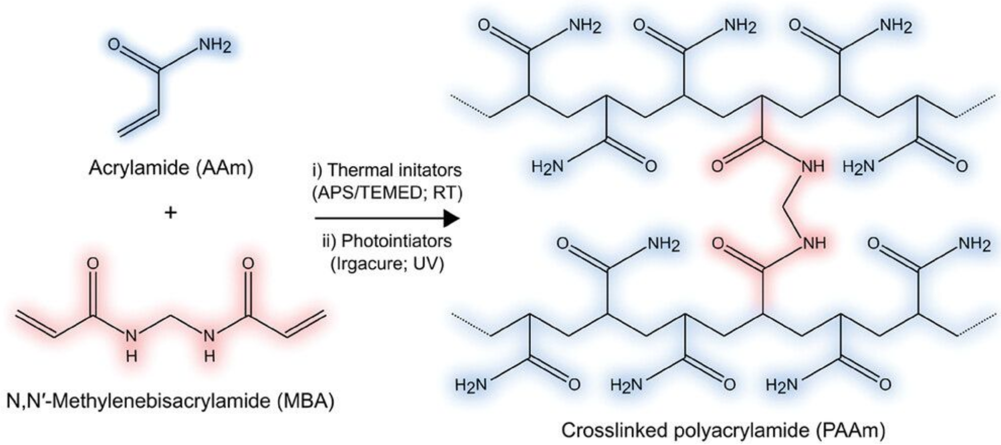


ACS Macro Lett. **2024**, *13*, 1369

Hydrogels

Synthetic Hydrogels

Polyacrylamide (PAAm): The Model Hydrogel



ADVANCED MATERIALS INTERFACES
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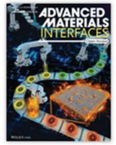

Volume 11, Issue 34
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Advanced Materials Interfaces Editors' Choice

Review | Open Access

Polyacrylamide Hydrogels as Versatile Biomimetic Platforms to Study Cell-Materials Interactions

Frano Milos, Aránzazu del Campo

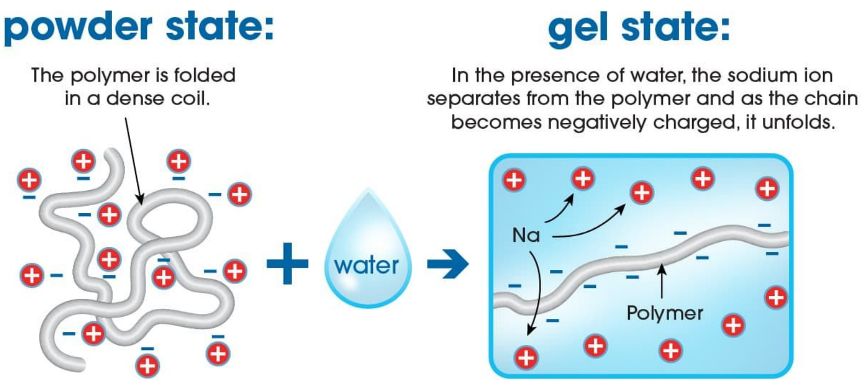
Abstract

Polyacrylamide (PAAm) hydrogels are widely adopted as 2D-model soft substrates for investigating cell-material interactions in a controlled in vitro environment. They offer facile synthesis, tunable physico-chemical properties, diverse biofunctionalization routes, optical transparency, mouldability in a range of geometries and shapes, and compatibility with living cells. PAAm hydrogels can be engineered to reconstruct physiologically relevant biointerfaces, like cell-matrix or cell–cell interfaces, featuring biochemical, mechanical, and topographical cues present in the extracellular environment. This Review provides a materials science perspective on PAAm material properties, fabrication, and modification strategies relevant to cell studies, highlighting their versatility and potential to address a wide range of biological questions. Current routes are presented to integrate cell-instructive features, such as 2D patterns, 2.5D surface topographies, or mechanical stiffness gradients. Finally, the recent advances are emphasized toward dynamic PAAm hydrogels with on-demand control over hydrogel properties as well as electrically conductive PAAm hydrogels for bioelectronics.

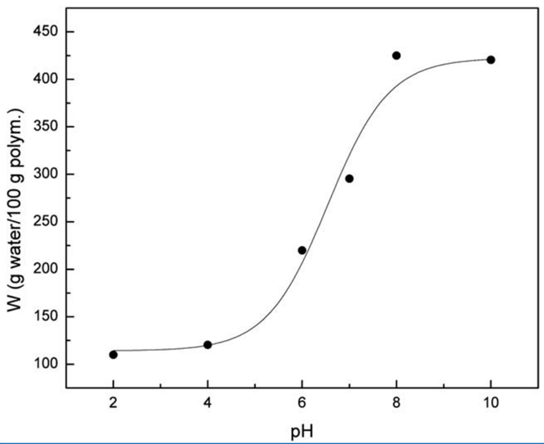
Hydrogels

Synthetic Hydrogels

Poly(acrylic acid): pH-Responsive Hydrogels



Soft Matter **2011**, *7*, 5847

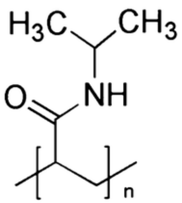


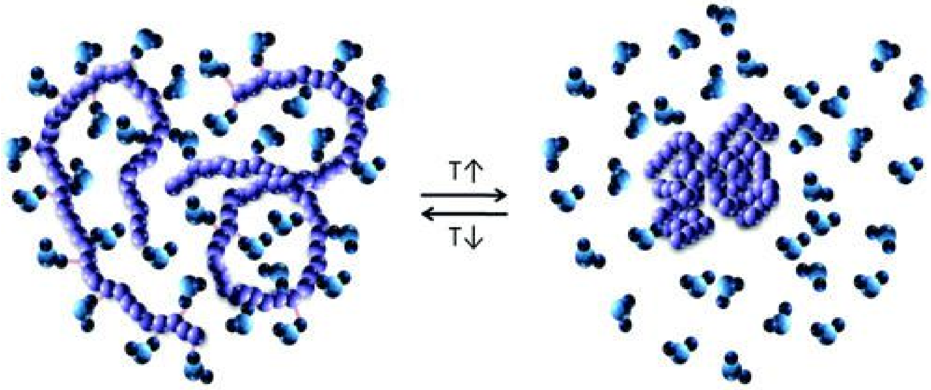
https://gelok.com/super-absorbent-polymer-overview/

Hydrogels

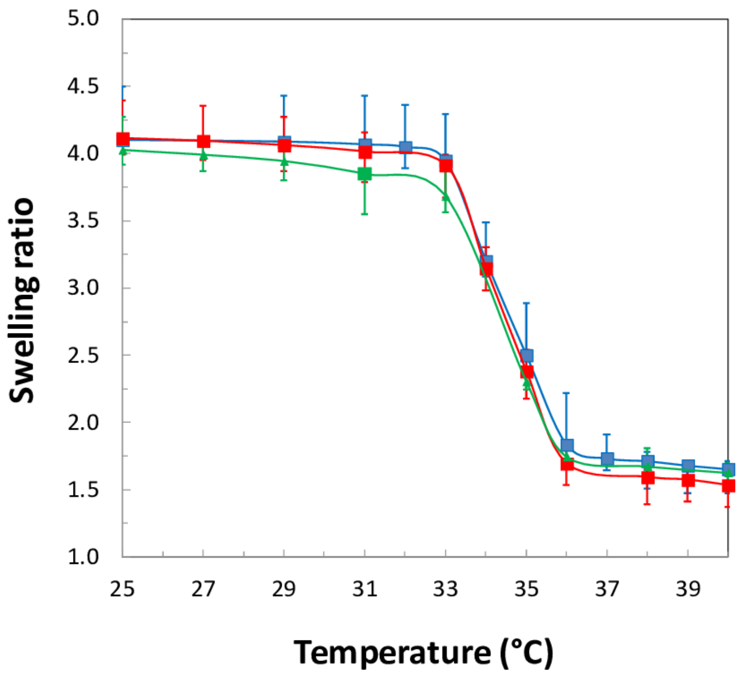
Synthetic Hydrogels

PNIPAM: Thermoresponsive Hydrogel (Case Study)





coil **globule**



Swelling ratio of PNIPAM hydrogel films as function of temperature for various dry thickness of the films (150 nm, 250 nm, 420 nm), using end-functionalized PNIPAM with the molecular weight equal to 254 kg/mol.

https://assignmentpoint.com/coil-globule-transition/

ACS Appl. Mater. Interfaces **2016**, *8*, 11729

Hydrogels

Network Architecture

Identical polymers can produce vastly different materials depending on network topology.

"Architecture" refers to the spatial arrangement of polymer chains and crosslinks, not the molecular structure of the repeat unit:

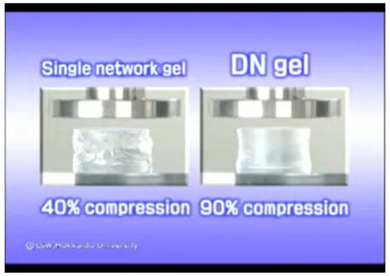
Examples:

- Homogeneous network
- Heterogeneous network
- Gradient network
- Double network
- Interpenetrating polymer network (IPN)
- Composite network (*e.g.*, with nanoparticles)

Polymers **2021**, *13*, 4259

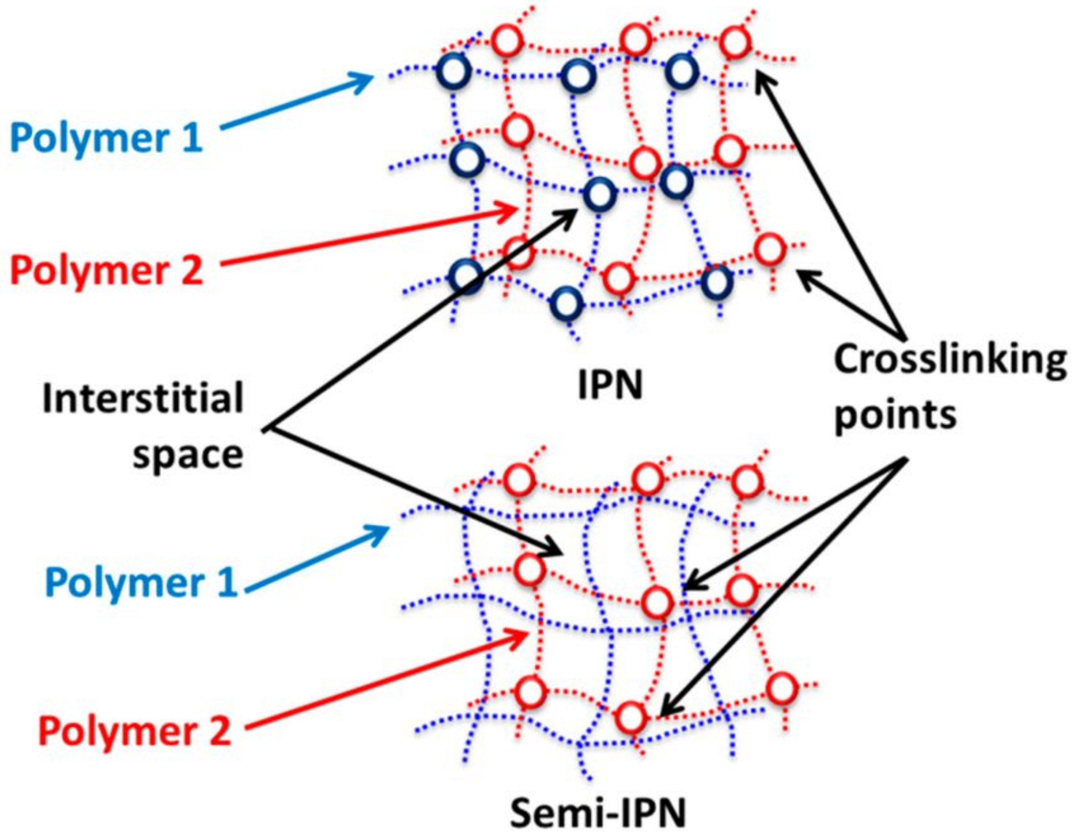
Hydrogels

https://youtu.be/5O5jB-EIHto?si=CyKl8T1fhZA4hwLv



Network Architecture

(Semi-)interpenetrating Networks



Polymers **2021**, *13*, 4259

IPNs

Combining two independently crosslinked networks allows **synergistic integration of complementary properties**, often resulting in improved mechanical strength, toughness, swelling control, and multifunctionality compared to a single-network hydrogel.

Semi-IPNs

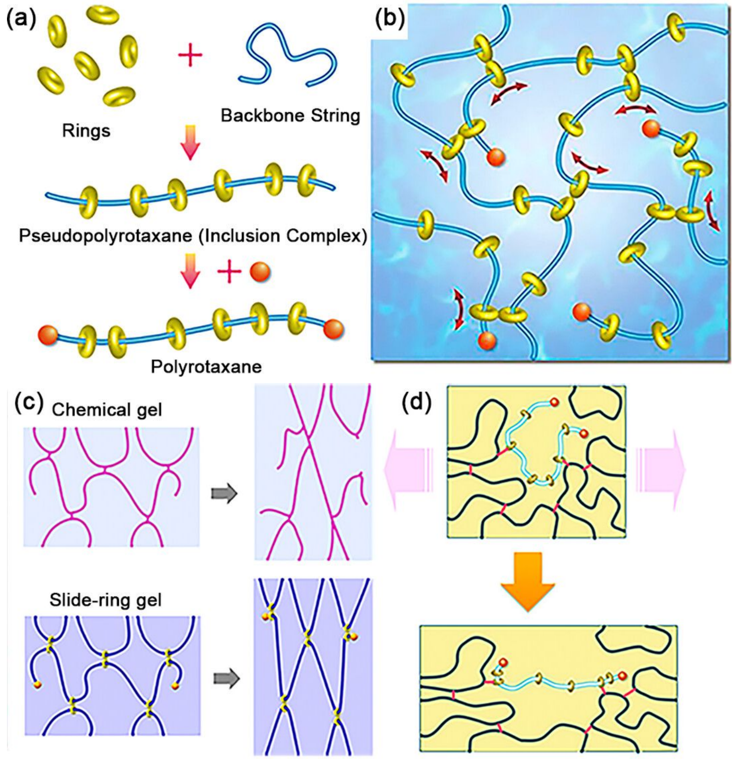
Incorporating a linear polymer into a crosslinked network enables the **combination of functionalities** (e.g., bioactivity, conductivity, responsiveness) without significantly altering the primary network structure or its synthesis.

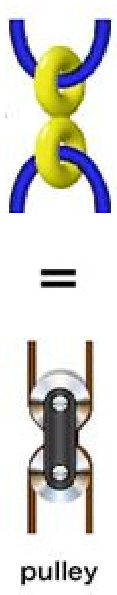
Hydrogels

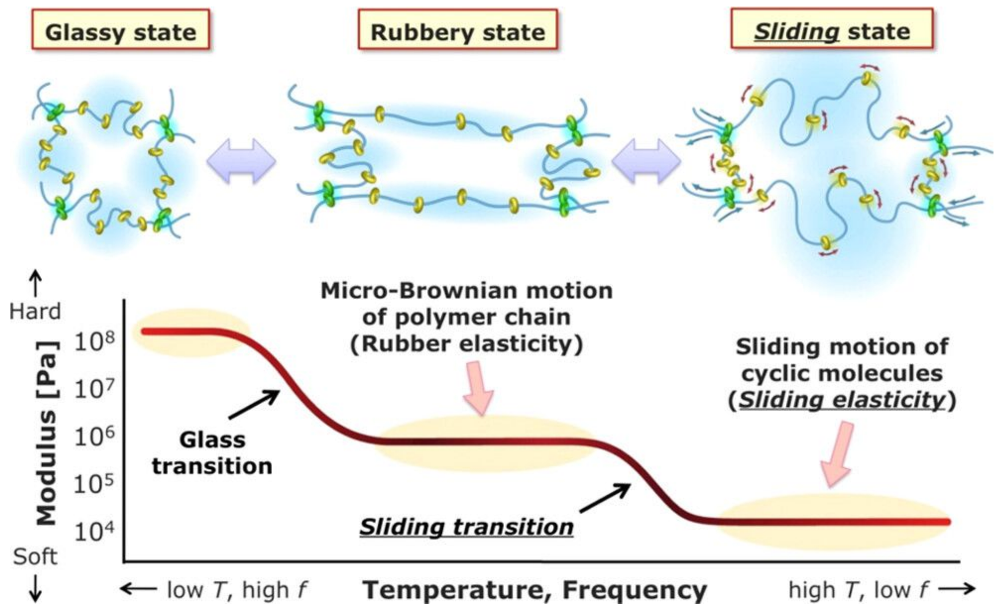
Network Architecture

Slide-Ring Gels

Slide-ring gels employ movable (topological) crosslinks that redistribute stress during deformation, resulting in exceptionally stretchable and fatigue-resistant hydrogel networks.







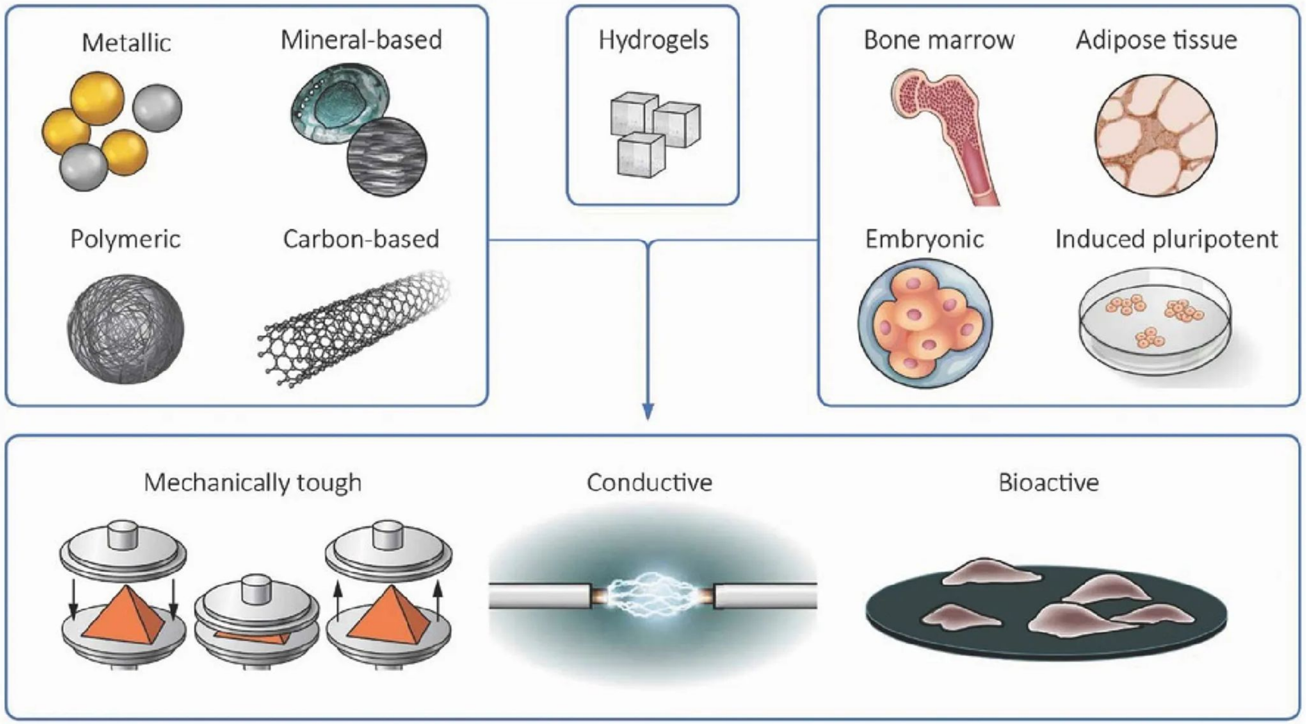
Macromolecules **2025**, *58*, 2157

Hydrogels

Composite Hydrogels

Incorporating a second material into the hydrogel network

e.g., silica nanoparticles, Laponite platelets, cellulose nanocrystals, graphene oxide sheets



Front. Bioeng. Biotechnol. **2020**, *8*, 611

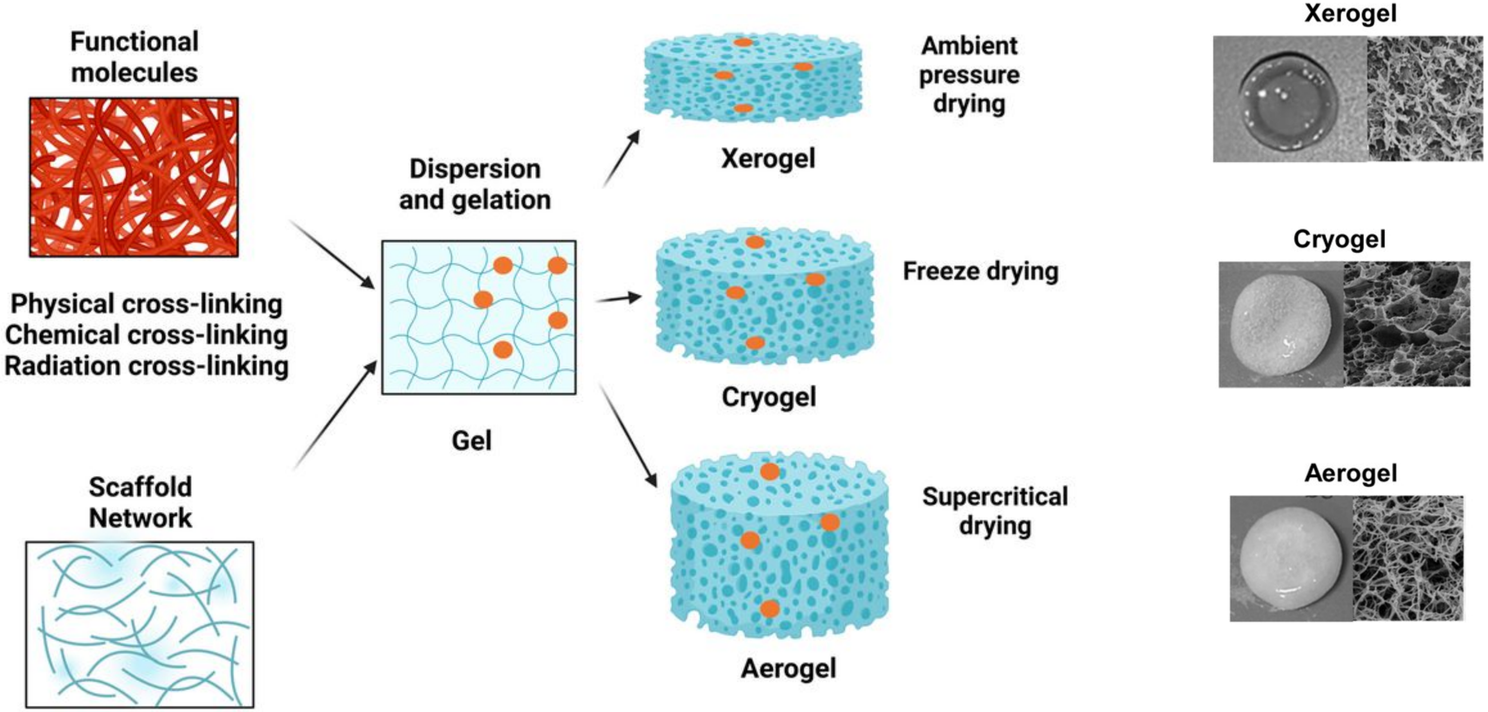
Hydrogels

Drying

Xerogel, Cryogel, Aerogel…

"Dry gels possess high porosity, large surface area, and tunable internal structures, enabling improved drug solubility, controlled release, and enhanced bioavailability."

Adv. Healthcare Mater. **2025**, ***14***, e00863



Safety **2023**, *9*, 80

Hydrogels

Summary

- Chemistry determines what interactions are possible.

- Architecture determines how those interactions are organized in space.

- Both must be considered before discussing the influence of size (hydrogels, microgels, nanogels).

Safety **2023**, *9*, 80